

C-STGB: Conformal Spatio-Temporal GraphBoost End-to-End Surveillance Platform

STAGE 1: STREAMING INGEST

DuckDB Invariants & Hawkes

Continuous Ingest & Subgraphs

$$G_t = (V_t, E_t), \quad N_K(u) \ (K \leq 15)$$

DuckDB / Arrow zero-copy streaming

Mass Flow Conservation

$$\Phi_{\text{flow}} = \log(1 + A_{\text{out}}/[A_{\text{in}} + \varepsilon]) \approx 1.0$$

Layering dissipation & pass-through detector

Hawkes Intensity & Causal Taint

$$\lambda_u(t) = \mu_u + \sum \alpha \cdot e^{-\beta \Delta t}$$

Forward & backward taint (< 0.08 ms)

Out: $\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}, \ G_t^{(K)}$

0.45 ms SLA | DuckDB Ingest

STAGE 2: GNN & FILTER

Tri-Band & Latent GraphSMOTE

Tri-Band Temporal Attention

$$\Phi_{\text{time}}(\Delta t), \ w(\Delta t) = \sum \pi_b e^{-\lambda_b \Delta t}$$

Microsecond bursts to 90-day dormancy

Learnable Edge-Trust Gating

$$\hat{g}_{ij} = \delta_{\text{floor}} + (1 - \delta_{\text{floor}})g_{ij}$$

Prunes 65.9% of adversarial chaff links

Typology Latent GraphSMOTE

$$h_{\text{syn}} = (1 - \rho)h_u + \rho h_v, \ u, v \in C_k$$

Latent interpolation & hard-negative pairs

Out: $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$ (Latent Embedding)

Continuous Temporal Dynamics

STAGE 3: FUSION & RISK

Evidence-Adaptive Bayes Head

Evidence-Adaptive Fusion Gate

$$\alpha_u = \sigma(\mathbf{w}_\alpha^\top [\mathbf{h}_u \parallel \mathbf{z}_{\text{inv}}] + b_\alpha)$$

Cold-start: $\text{deg}(u) \leq 2$ shifts to $\mathbf{z}_{\text{inv}}$

Unified Representation Space

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u + (1 - \alpha_u)[\mathbf{z}_{\text{inv}} \parallel \mathbf{x}_u]$$

Secures 89.6% F1 on isolated entities

Cost-Sensitive Bayes Head

$$\tau^* = \arg \min_{\tau} (15 \cdot \text{FN} + 1 \cdot \text{FP})$$

Asymmetric loss policy (C_FN / C_FP = 15)

Out: $\hat{p}_u \in [0, 1], \ \mathbf{h}_u^* \in \mathbb{R}^{d^*}$

Cold-Start Guarded + Bayes Risk

STAGE 4: CONFORMAL SWARM

CRC Triage & FinCEN SAR

Class-Conditional CRC

$$q^{(y)} = \text{Quantile}(D_{\text{cal}}), \ 1 - \alpha \geq 99.0\%$$

Finite-sample coverage with ACI tracking

3-Tier Deterministic Triage

$$\Gamma(u) \in \{\{1\}, \{0, 1\}, \{0\}\}$$

Routes >99.4% to straight-through tiers

Multi-Agent Forensic Swarm

Investigator $\rightarrow$ Auditor $\rightarrow$ Drafter

FinCEN Form 111 XML + SHA-256 Merkle

Out: $\Gamma(u) \subseteq \{0, 1\}, \ \text{XML}_{\text{SAR}}$

Model Risk Governance (SR 26-2)